MIL-L-36142(BuMed)
15 July 1963

MILITARY SPECIFICATION

LITTER EXTENSION, PATIENT'S PILLOW SUPPORT-BACKREST

1. SCOPE

1.1. This specification covers a litter extension combination to increase the usable space of a litter, and to provide a pillow support-backrest for those patients required to sit upright when transported by aircraft.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon.
QQ-Z-325	Zinc-Coating, Electrodeposited, Requirements for.
CCC-C-419	Cloth, Cotton, Duck, Unbleached, Plied-Yarns (Army and Numbered).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.

Military

JAN-E-480	Enamel, Baking, Phenol-Or Urea-Formaldehyde.
MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-B-543	Buckles, Tongueless and Web Strap.
MIL-W-4088	Webbing, Textile, Woven Nylon.
MIL-T-5038	Tape, Textile, Reinforcing, Nylon.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6530

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods.
Fed. Std. No. 751	Stitches, Seams, and Stitchings.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Medical Supply Center

20571	Litter Extension, Patient's Pillow Support-Backrest.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Materials. Except as herein specified, materials shall conform to the applicable portions of the specifications, standards, and drawing listed in section 2, and shall be capable of meeting the requirements shown therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.4.1. When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

3.1.1 Cloth, cotton, duck. Cloth for the backrest shall conform to No. 10, hard texture, type I of Specification CCC-C-419. The cloth shall be dyed olive drab shade No. 7.

3.1.2 Webbing.

3.1.2.1 Webbing, cotton. The webbing shall be dyed olive drab No. 7 or approximate the shade of the cotton duck, and shall conform to Specification MIL-W-530, as follows:

For the pole hood- Type II, class 4, 3-inch.
For securing straps- Type III, class 4, 1-inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

3.1.2.2 Webbing, nylon. The webbing sewn to the two sides of the duck panel and the litter hook assembly shall be dyed olive drab No. 7, or approximate the shade of the cotton duck, and shall conform to type VIII, 1-23/32-inch of Specification MIL-W-4088.

3.1.3 Tape, nylon. Tape shall conform to Specification MIL-T-5038, shall be dyed olive drab No. 7, or approximate the shade of the cloth duck, and shall conform to the following:

For binding the bottom and side edges of the duck panel	Type III, 3/4-inch
For the lattice	Type II, 1-inch

3.1.4 Thread, nylon. Thread shall conform to Specification V-T-295, shall be dyed olive drab No. 7, or approximate the shade of the cloth duck, and shall conform to the following:

For all seams, stitchings, and cross box and box stitching, except the cross box stitching joining webbing pieces to the duck panel adjacent to the pole hood	Type I, class 1, letter size E
For cross-bar stitching joining webbing pieces to duck panel adjacent to the pole hood	Type II, number size 5

3.1.5 Litter hook and loop.

3.1.5.1 Litter hook. The litter hook shall be fabricated from 0.062 ± 0.003 by 1.500 ± 0.005 inch, steel FS 1050 carbon steel hardened to spring temper and constructed in accordance with the drawing. The finish shall be as specified in 3.1.7.

3.1.5.2 Litter loop. The litter loop which is a component of the litter hook shall be fabricated from steel wire, 0.156 ± 0.002 diameter, 1/4 hard, and constructed in accordance with the drawing. The finish shall be as specified in 3.1.7.

3.1.6 Buckle. The buckle for the retaining straps shall be double bar with slide conforming to type III, style 3, construction A, class 3, size 1-inch of Specification MIL-B-543. The finish shall be as specified in 3.1.7.

3.1.7 Finish. All the metal parts shall be given a finish consisting of a phosphate treated zinc plate conforming to type III, class 2 of Specification QQ-Z-325, followed by bake enameling. The enamel shall conform to type II of Specification JAN-E-480. Color shall be black.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

3.2 Construction. The litter extension shall be constructed as shown in the drawing, and as specified herein.

3.2.1 Canvas body. The canvas body shall be fabricated from one piece of canvas as specified in 3.1.1. The free end and side edges shall be bound with nylon tape (binding); the opposite end shall have a 1-5/8 inch finished hem with the raw edge turned under 1/2 inch. In addition, a length of nylon webbing shall be stitched along the length of each side. There shall be a litter hook assembly attached to the canvas piece on the free end, in two places, as shown in the drawing. A 3- by 4-inch canvas (pole) hood shall be attached to each side edge of the body with the opening of the hood facing the opposite end, as shown on the drawing.

3.2.2 Lattice portion. The lattice portion as shown in the drawing shall consist of five equal lengths of nylon webbing with one end of each stitched to the canvas body, and three equal lengths of nylon webbing stitched across (perpendicular to) the five nylon webbing lengths. These webbing lengths shall be interlaced as shown in the drawing, and where overlapped, stitched together with a 3/4-inch box-stitch. The free end (top) of the lattice section shall consist of a doubled length of nylon webbing within which the ends of the five nylon webbing lengths are sandwiched and stitched together.

3.2.3 Attachment of securing straps. The securing straps shall be attached to the litter extension as shown in the drawing. At the point of attachment of the securing straps to the body, the loops for the buckles shall be 4-1/2 ± 1/4 inches overall length with the raw edge turned under 1-1/2 ± 1/8 inches. The cotton webbing shall be threaded over the center bar of a 1-inch double-bar buckle; a 3/4-by 1-inch cross-box stitch shall be sewn through all the plies of materials.

3.3 Seams, stitchings, and stitches. Seams, stitchings, and stitches shall conform to Federal Standard No. 751, as follows:

For all seams, stitchings, box and cross-box stitchings, except for the joining of securing straps to the duck panel adjacent to the pole hood.	Type 301, 8 to 10 stitches per inch.
For the cross-box stitching joining the securing straps to duck panel adjacent to the pole hood.	Type 301, 4 to 6 stitches per inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

3.3.1 Box and cross-box stitchings. Unless otherwise specified, the dimensions of the box and cross-box stitchings for attaching the nylon tapes to one another, and the cotton webbing to the nylon tapes, shall be 3/4 by 3/4 inch box-stitch.

3.4 All cut ends of the nylon webbing pieces shall be fused. The free end of the securing straps shall be impregnated with either cellulose acetate or cellulose acetate butyrate for a distance of 1 ± 1/4 inches to prevent fraying.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities, or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105.

	Inspection level	AQL (percent defective)	Minimum sample size
Component material testing	L2	*	3

* Acceptance number shall be zero (0) for all sample sizes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

4.2.3 For chemical analysis of metals. Four ounces of each lot of each metal used in the construction of the litter extension shall be selected for chemical analysis as specified in 4.4.1.

4.2.4 Component materials. Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the litter extension to perform tests specified herein. When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.2.2. The unit of product for testing shall be the quantity necessary to perform all required tests one time each, taking into consideration the fact that, where possible, the same material may be used to perform two or more tests. Lot sizes are expressed in multiples of the following:

Duck	Yards or fractions of a yard.
Webbing, cotton	Yards or fractions of a yard.
Webbing, nylon	Yards or fractions of a yard.
Tape, nylon	Yards or fractions of a yard.
Thread, nylon	Cone or spool.
Litter hook	Each.
Litter loop	Each.
Buckle	Each.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The litter extension shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT	CLASSIFICATION MAJOR	MINOR
Fabric	Holes, cuts, or tears.	X	
	Other fabric defects;		
	Seriously affecting serviceability	X	
	Affecting serviceability but not too serious		X
Webbing (all types)	Frayed or scalloped edges, not firmly or tightly woven	X	
	Type and color not as specified		X
	Width not as specified (other than for buckle)		X
	Width of webbing does not fit freely through buckle opening impairing not serving intended purpose	X	
Hardware	Broken or malformed, failing to serve intended purpose	X	
	Finish partly or totally omitted		X
	Burrs or sharp edges which may damage material or cause injury in handling	X	
	Wrong type or design not serving intended purpose	X	
	Wrong shade		X
Buckle	Fails to function	X	
Seams and Stitching			
Open seams	On canvas body-		
	-more than 1/2 inch	X	
	-1/2 inch or less		X
	On lattice (except X-pieces)		
	-more than 3/4 inch	X	
	-3/4 inch or less		X

Note: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more skipped stitches or runoffs occur.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT	CLASSIFICATION MAJOR	CLASSIFICATION MINOR
Runoffs (see Open seams)			
Raw edges	When securely caught in stitching more than 3/4 inch	X	

Note: Raw edges not securely caught in stitching shall be classified as open seams.

Stitch margins-gage	Not as specified, affecting serviceability		X
Stitching ends	Backstitched less than 3/8 inch on canvas body or webbing (except where held down by other stitching or turned under in a hem)		X
Thread breaks.	Backstitched less than 3/4 inch		X

Note: Thread breaks not backstitched as required shall be considered as open seams.

Seam and stitch type	Wrong seam or stitch type	X	
Stitch tension	Loose, resulting in loosely exposed bobbin or top thread; tight resulting in tightness of fabric, webbing, or seam		X
Stitches per inch	One stitch less than minimum or maximum specified		X
	Two or more stitches less than minimum or maximum specified	X	

Note: Variations in the number of stitches per inch caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners shall be classified as follows:

Within the minor classification
-no defect
Within the major classification
-minor defect

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

EXAMINE	DEFECT	CLASSIFICATION	
		MAJOR	MINOR
Component and assembly	Any required component or operation omitted (unless otherwise classified herein)	X	
Stitchings	Needle chews	X	
	Seams badly puckered or pleated		X
	Mends, darns, or patches		X
	Cross box stitching less than specified by more than 1/8 inch		X
Hem	Less than specified width by more than 1/8 inch		X
Binding	Omitted	X	
Buckle	One or both omitted	X	
	(Tooth) end of buckle not facing lattice, not serving intended purpose	X	
Litter hook and loop	Not assembled to canvas body as specified not serving intended purpose	X	
Webbing, cotton	Ends of securing straps not impregnated with plastic as specified		X
Lattice	Not constructed as specified	X	
Canvas Body	Not constructed as specified	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis of the metals. The samples selected in accordance with 4.2.3 shall be analyzed in conformance with methods 111 or 112 of Federal Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) litter extension, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be neatly folded and packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-636 or PPP-B-676. Commercial colors shall be acceptable when carton conforming to Specification PPP-B-676 is utilized. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP-B-636. Closure shall be adequate to prevent spillage of contents under normal handling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

5.2 Packing

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood, PPP-B-621, table III; or PPP-B-636, class 2

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547 Closure and sealing shall conform to applicable paragraphs of appendix thereto

5.2.1.3 Strapping Strapping, when required, shall be in accordance with applicable paragraphs of appendix thereto

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container Exterior container shall conform to Specification PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with Standard MIL-STD-129 When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129 In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated Service color markings are required only on direct shipments from the contractor's plant to overseas activities.

6 NOTES

6.1 Ordering data Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification,

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.3 Transportation description. Transportation description applicable to this item is:

Dry Goods, NOI, in boxes

Carload minimum weight, 20,000 pounds subject to Rule 34 of the Uniform Freight Classification

Truckload minimum weight, any quantity

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army. - MD
Navy. - BuMed
Air Force. - AFCSC

Preparing activity:
Defense Supply Agency. - DM

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of the specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION Mil Spec. MIL-L-36142(BuMed), dated 15 July 1963 - Litter Extension, Patient's Pillow Support-Backrest

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A
☐ Direct Government Contract ☐ Subcontract

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.

FOLD

Defense Supply Agency
Defense Medical Supply Center
Third Avenue and 29th Street
Brooklyn, N.Y., 11232

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Defense Supply Agency
Defense Medical Supply Center
Third Avenue and 29th Street
Brooklyn, XXX N. Y., 11232
Attn: Technical Dept., T-M2

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:07Z
Check the source to verify that this is the current version before use.